AI-Driven Academic Risk & Dropout Early-Warning System

Presented by

Altrin Benser A - 962223205010

Jerish Rethna J - 962223205029

John Jins J - 962223205032

Mukhilan S - 962223205040

Guide:

Er. Agnes Alex Rathy P

Assistant Professor/IT

Abstract, Problem Statement & Objectives

Abstract

- Traditional systems find weak students only after exam results and keep data as static records.
- Digital Class Twin captures attendance, assessment, quiz and behavior data as events and keeps a live digital twin of each student.
- Risk is scored with explainable factors and shown on role-based dashboards for teachers, counselors and parents.

Problem Statement

- Weak students are identified only after examination or semester results.
- Static records do not show how performance changes over time.
- Attendance-threshold rules miss the complete academic situation.
- No mechanism for continuous monitoring and early intervention.

Objectives

1. Continuously collect attendance, assessment and behavior data
2. Maintain a Digital Twin per student: current state, history, sync status
3. Predict academic risk: Low, Medium, High
4. Predict future GPA using machine learning (planned)
5. Analyze trajectory: Declining, Stable, Recovering
6. Give explainable factors behind every prediction
7. Provide what-if simulation for interventions
8. Support early intervention by recording actions and outcomes

Related Sustainable Development Goals (SDG)

PRIMARY SDG

SDG 4: Quality Education

SDG 4

Closest match to our project objective

Early identification of academically at-risk students for timely support

- Risk score from academic performance, attendance and behavior
- Low / Moderate / High levels show who needs help first
- Dashboards let teachers and parents act early
- Aims to improve educational outcomes and reduce dropout

SECONDARY SDG

SDG 10: Reduced Inequalities

SDG 10

- Early identification can help students who need extra academic support
- Supports more equitable access to timely guidance
- A supporting link, not a claim to remove inequality

SECONDARY SDG

SDG 9: Industry, Innovation & Infrastructure

SDG 9

- Applies digital-twin concepts to student academic profiles
- Data-driven analytics and digital dashboards in education
- Rule-based risk engine today; trained ML planned

The project primarily contributes to SDG 4 by supporting early identification of academic risk and timely educational intervention.

Literature Survey , Summary and Research Gap

Research Gap

1. Many studies use static or institution-specific datasets.
2. Most focus mainly on prediction rather than continuous monitoring.
3. Limited integration of academic, attendance and behavioural events.
4. Few systems connect predictions directly with interventions.

Major Methods

1. Machine Learning: Random Forest, Decision Tree, SVM, Logistic Regression
2. Ensemble & Boosting methods for dropout prediction
3. Educational Data Mining and Learning Analytics
4. SHAP / Explainable AI for prediction interpretation
5. Digital Twin & graph-based student pathway modelling

Key Findings

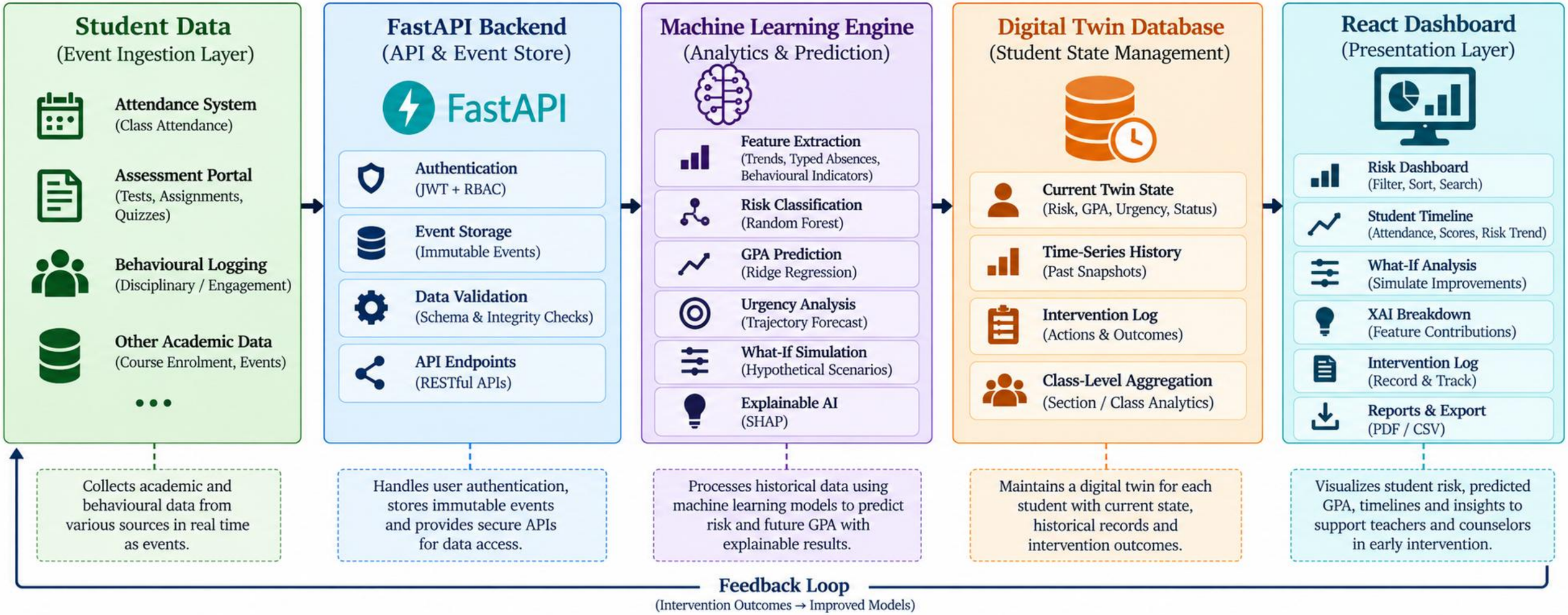
1. ML can predict academic performance and dropout risk at an early stage.
2. Attendance, academic performance, LMS activity and behaviour are important indicators.
3. Combining multiple student data sources improves risk prediction.
4. XAI helps identify the factors influencing predictions.

Relevance to Our Project

1. Our Digital Class Twin extends existing ML and Digital Twin approaches.
2. Uses event-driven continuous monitoring and time-series student state.
3. Provides Low/Medium/High risk, GPA prediction and trajectory-based urgency.
4. Uses SHAP, What-If simulation and intervention logging for actionable decisions.

Proposed System Architecture

Events → FastAPI ingestion → Rules & features → Risk engine → Twin & alerts → Database → REST API (JWT) → React dashboards



Design Methodology (Model, Algorithm, DB Design)

Algorithm

1. Real Time Events Ingestion(attendance, marks, behavior)
2. Normalize, validate and store event
3. Feature Extraction: attendance %, marks trend, behavior
4. Predictive Modeling (Ridge Regression & Random Forest): Apply Ridge Regression to forecast continuous GPA trajectory and Random Forest classification to evaluate historical dropout likelihood.
5. Explainable AI (XAI) & Hysteresis State Machine - update states (Stable, Declining, Critical)
6. Raise alert on risk change, with explanation

Model

Composite risk score

$$score = 0.5 \times Academic + 0.3 \times Attendance + 0.2 \times Behavior$$

- ≥ 75 Low → stable
- 50 - 74 Moderate → warning
- < 50 High → critical

Rule/threshold-based, explainable model (risk_engine.py). Trained ML model is the next step.

DB Design

Users & access
User, UserSession, ParentChildMap, TeacherClassMap

Events & features
Event, StudentFeatures

Risk & twin
RiskScore, DigitalTwin, TwinSnapshot, Alert, InsightReview

SQLAlchemy ORM · SQLite (dev) · Alembic migrations

Implementation Details (Framework, Design)

Framework / Technology Stack

Layer	Technologies
Frontend	React, Vite, Axios, React Router, custom CSS
Backend	Python, FastAPI, Uvicorn, Pydantic
Database	SQLAlchemy ORM (SQLite dev), Alembic migrations
Security	JWT authentication, password hashing, role-based access
ML engine	Python; rule-based scoring (risk_engine.py, forecasting_engine.py)
Tools	Git/GitHub, Swagger (/docs), Postman

Project modules

- backend/ - FastAPI APIs, auth, event pipeline, DB models
- client-app/ - React + Vite role-based dashboards
- ml-engine/ - features, risk, forecasting, twin updater
- shared-contracts/ - event and twin JSON schemas

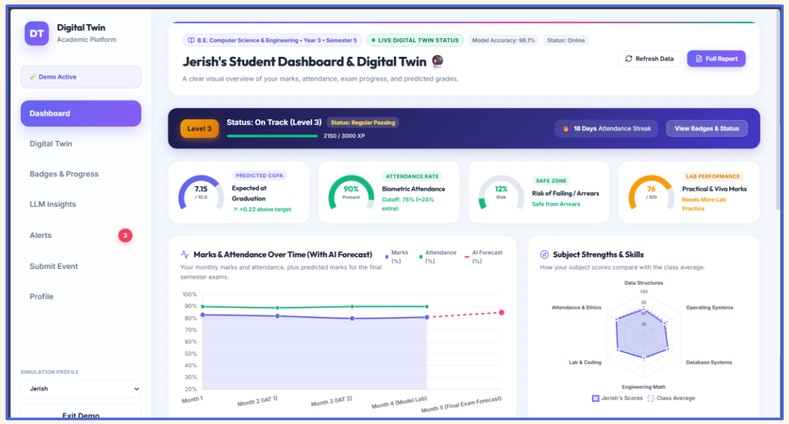
Role-based dashboards

- Student: Dashboard, Digital Twin, Insights, Alerts, Submit Event, Rewards, Profile
- Teacher: Dashboard, Class Overview, Risk Hub
- Parent: Dashboard, Child Twin, Academic Reports
- Admin: Dashboard, Audit Log, System Health

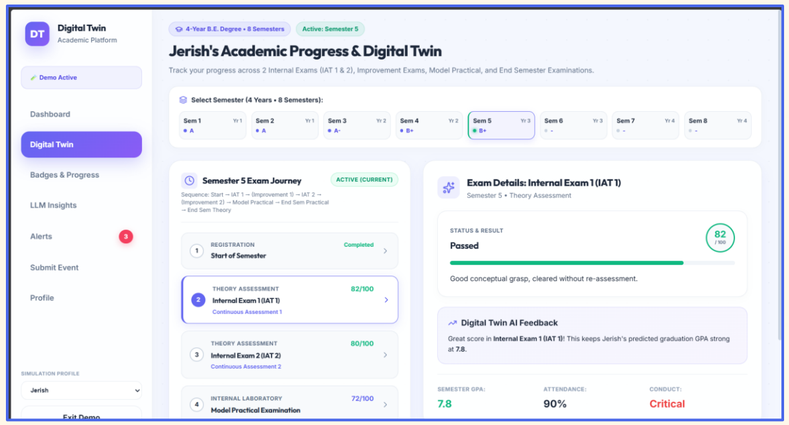
REST API routers

- auth, events, twins, insights, alerts, dashboard, profile, sessions, admin

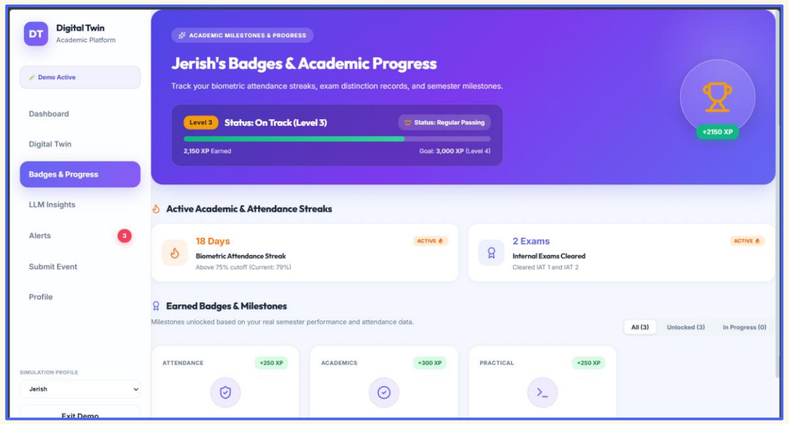
Output Screenshots(Desktop)



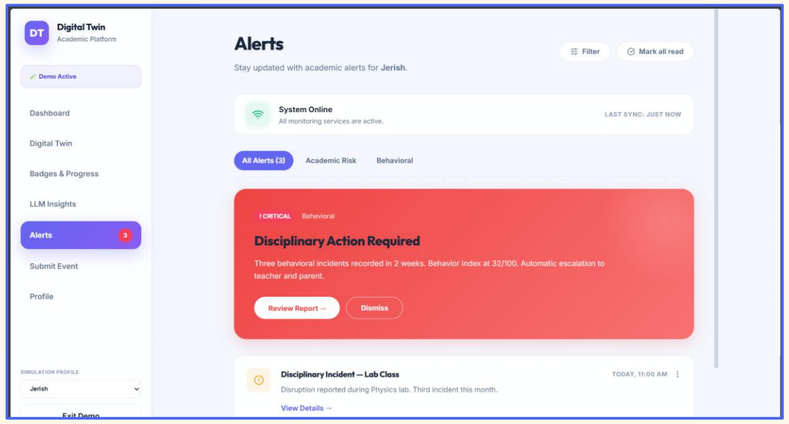
1. Student Dashboard



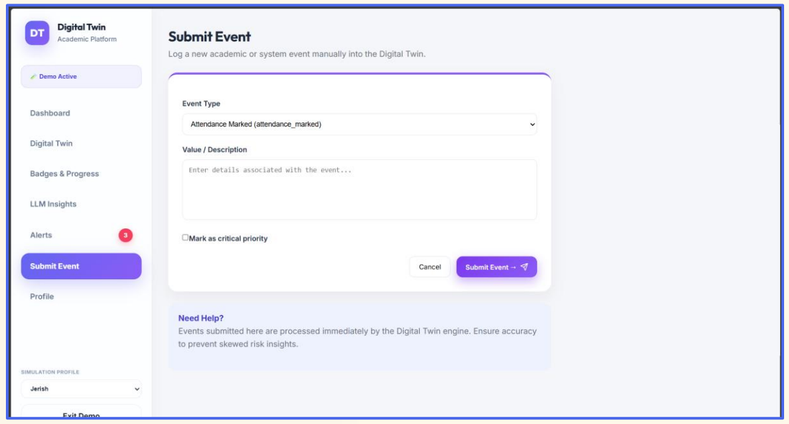
2. Digital Twin - semester progress



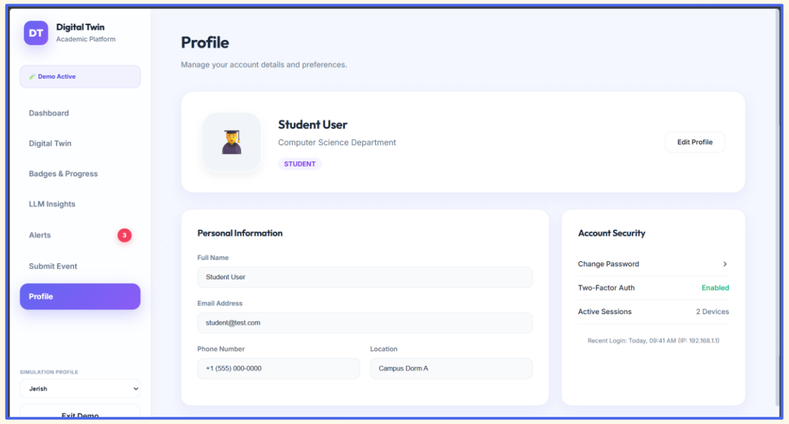
3. Badges & Progress



4. Alerts



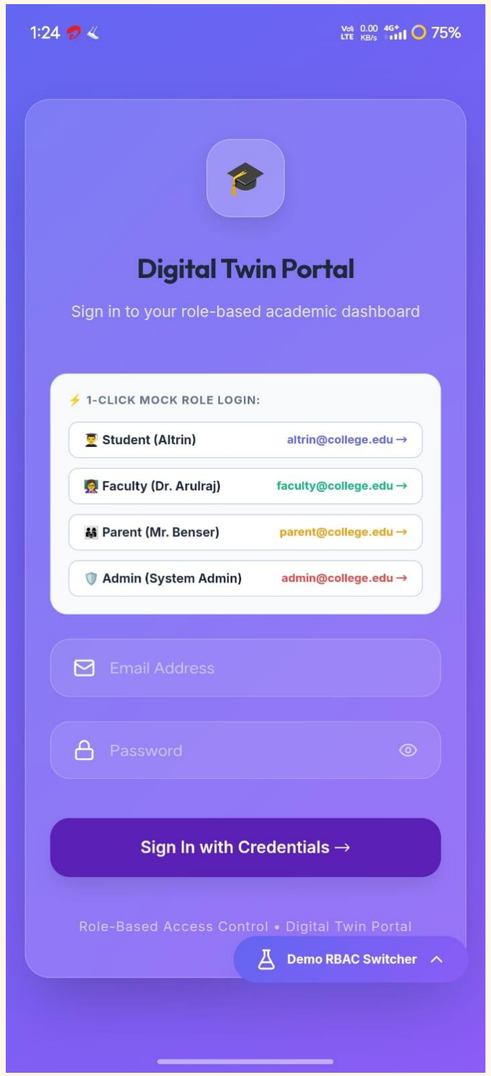
5. Submit Event

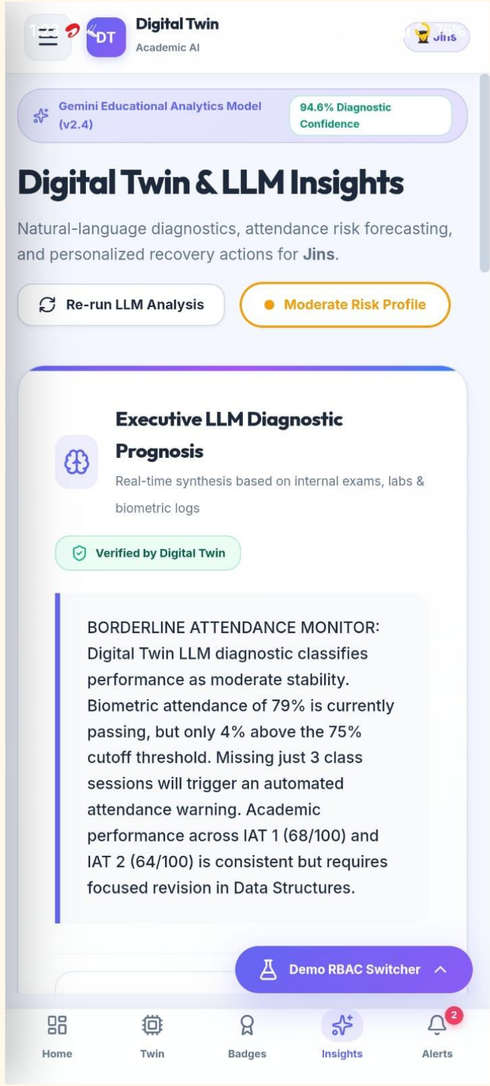


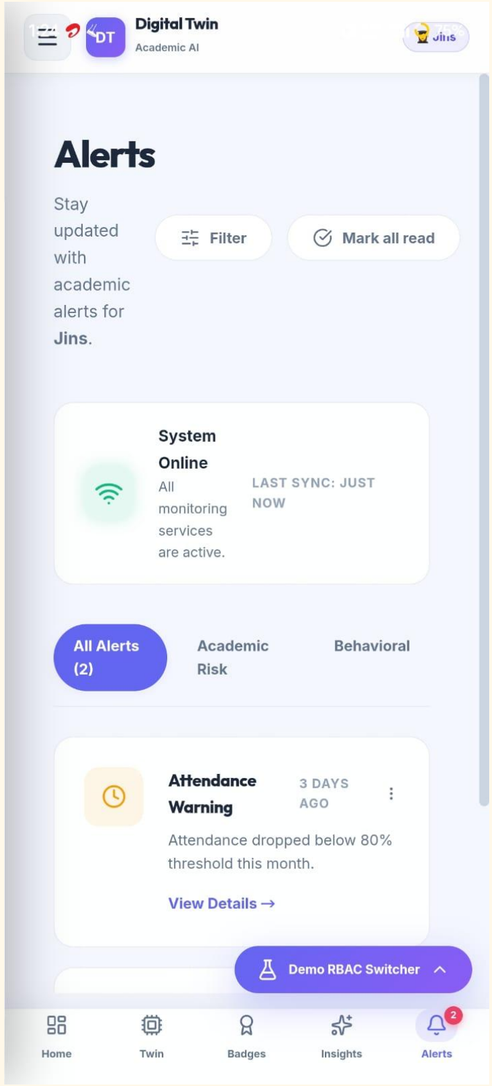
6. Profile

Screenshots from the running prototype in demo / simulation mode.

Output Screenshots(Mobile)







Conclusion

Conclusion

- Working end-to-end prototype: events → features → risk → Digital Twin → alerts → dashboards
- Explainable risk scores with twin snapshots and trend history
- JWT login with role-based access for student, teacher, parent and admin
- Risk engine is currently rule-based

Future Work

- Trained ML model for risk prediction
- What-if simulator (shadow twin) for interventions
- Closed-loop calibration against real outcomes
- PostgreSQL database and deployment

References

1. Cagliero et al. (2021). Predicting Student Academic Performance by Means of Associative Classification.
2. Karalar et al. (2021). Predicting Students at Risk of Academic Failure Using Ensemble Model During Pandemic in a Distance Learning System.
3. Park & Yoo (2021). Early Dropout Prediction in Online Learning of University Using Machine Learning.
4. Dabhade et al. (2021). Educational Data Mining for Predicting Students' Academic Performance Using Machine Learning Algorithms.
5. Tamada et al. (2022). Predicting Students at Risk of Dropout in Technical Course Using LMS Logs.
6. Yağcı (2022). Educational Data Mining: Prediction of Students' Academic Performance Using Machine Learning Algorithms.
7. Oqaidi et al. (2022). Towards a Students' Dropout Prediction Model in Higher Education Institutions Using Machine Learning Algorithms.
8. Segura et al. (2022). Machine Learning Prediction of University Student Dropout: Does Preference Play a Key Role?
9. Katsuragi & Tanaka (2022). Dropout Prediction by Interpretable Machine Learning Model Towards Preventing Student Dropout.
10. Su et al. (2022). Applying Machine Learning Technologies to Explore Students' Learning Features and Performance Prediction.
1. Plak et al. (2022). Early Warning Systems for More Effective Student Counselling in Higher Education.
2. Qin et al. (2023). Early Warning of Student Performance With Integration of Subjective and Objective Elements.
3. Cho et al. (2023). A Study on Dropout Prediction for University Students Using Machine Learning.
4. Balcıoğlu & Artar (2023). Predicting Academic Performance of Students with Machine Learning.
5. Nagy & Molontay (2023). Interpretable Dropout Prediction: Towards XAI-Based Personalized Intervention.
6. Villar & Robledo Velini de Andrade (2024). Supervised Machine Learning Algorithms for Predicting Student Dropout and Academic Success: A Comparative Study.
7. Vaarma & Li (2024). Predicting Student Dropouts with Machine Learning: An Empirical Study in Finnish Higher Education.
8. Tiukhova et al. (2024). Explainable Learning Analytics: Assessing the Stability of Student Success Prediction Models by Means of Explainable AI.
9. Miranda et al. (2024). Machine Learning's Model-Agnostic Interpretability on the Prediction of Students' Academic Performance in Video-Conference-Assisted Online Learning.
10. Alalawi et al. (2024/2025). An Extended Learning Analytics Framework Integrating Machine Learning and Pedagogical Approaches for Student Performance Prediction and Intervention.

Participation in Technical Event

Planned Paper Submission at ICAESEM 2026
October 30-31, 2026